

Assignment 2

- 1.16.** Consider a discrete-time system with input $x[n]$ and output $y[n]$. The input-output relationship for this system is

$$y[n] = x[n]x[n-2].$$

- (a) Is the system memoryless?
- (b) Determine the output of the system when the input is $A\delta[n]$, where A is any real or complex number.
- (c) Is the system invertible?

- 1.17.** Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t)).$$

- (a) Is this system causal?
- (b) Is this system linear?

- 1.18.** Consider a discrete-time system with input $x[n]$ and output $y[n]$ related by

$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k],$$

where n_0 is a finite positive integer.

- (a) Is this system linear?
- (a) Is this system time-invariant?
- (c) If $x[n]$ is known to be bounded by a finite integer B (i.e., $|x[n]| < B$ for all n), it can be shown that $y[n]$ is bounded by a finite number C . We conclude that the given system is stable. Express C in terms of B and n_0 .

- 1.19.** For each of the following input-output relationships, determine whether the corresponding system is linear, time invariant or both.

- (a) $y(t) = t^2 x(t-1)$
- (b) $y[n] = x^2[n-2]$
- (c) $y[n] = x[n+1] - x[n-1]$
- (d) $y[n] = \mathcal{O}d\{x(t)\}$

- 1.20.** A continuous-time linear system S with input $x(t)$ and output $y(t)$ yields the following input-output pairs:

$$\begin{aligned} x(t) = e^{j2t} &\xrightarrow{S} y(t) = e^{j3t}, \\ x(t) = e^{-j2t} &\xrightarrow{S} y(t) = e^{-j3t}. \end{aligned}$$

- (a) If $x_1(t) = \cos(2t)$, determine the corresponding output $y_1(t)$ for system S .
- (b) If $x_2(t) = \cos(2(t - \frac{1}{2}))$, determine the corresponding output $y_2(t)$ for system S .

Assignment 2

1.30. Determine if each of the following systems is invertible. If it is, construct the inverse system. If it is not, find two input signals to the system that have the same output.

(a) $y(t) = x(t - 4)$

(b) $y(t) = \cos[x(t)]$

(c) $y[n] = nx[n]$

(d) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

(e) $y[n] = \begin{cases} x[n-1], & n \geq 1 \\ 0, & n = 0 \\ x[n], & n \leq -1 \end{cases}$

(f) $y[n] = x[n]x[n-1]$

(g) $y[n] = x[1-n]$

(h) $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau) d\tau$

(i) $y[n] = \sum_{k=-\infty}^n (\frac{1}{2})^{n-k} x[k]$

(j) $y(t) = \frac{d x(t)}{dt}$

(k) $y[n] = \begin{cases} x[n+1], & n \geq 0 \\ x[n], & n \leq -1 \end{cases}$

(l) $y(t) = x(2t)$

(m) $y[n] = x[2n]$

(n) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$